

QST NETWORK MANDATE

1. **Payload Extraction:** Read the telemetry file. For each row, Base64-decode the payload, decompress it using Bzip2 (bz2), and split the resulting string by the pipe | character to isolate the Raw Qubits (integer) and the State Token (string).
2. **Solar Interference Check:** Convert the Node's Unix Epoch Timestamp to UTC format. If the calendar **day of the month** is a prime number, you must immediately multiply the Raw Qubits by **0.5**. If it is not prime, the multiplier is 1.0.
3. **Topology Calculation:** Look at the Topological_Lattice.jpg DAG diagram. Count the absolute minimum number of hops from the 'Core' to the target Node. The Topology Multiplier is calculated as: $1.0 - (0.1 * \text{hops})$.
4. **Final Calculation:** Multiply the (potentially penalized) Raw Qubits by the Attenuation factor from the Excel file, and then multiply by the Topology Multiplier to achieve the Final Key Rate (bits).